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MTH632 Complex Analysis and Differential Geometry

Question No : 12 of 33

Let Z be a complex valued function then $\arcsin(Z) =$ _____.

Answer (Please select your correct option)

- $\log(iz + (1-z^2)^{1/2})$
- $-i\log(iz + (1-z^2)^{1/2})$
- $\log(iz - (1-z^2)^{1/2})$
- $-i\log(iz - (1-z^2))$

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MTH632 Complex Analysis and Differential Geometry

Question No : 30 of 33

Evaluate the integral: $\int_0^{\pi/6} e^{i2t}\,dt$.

Answer (Please click here to Add Answer)

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MTH632 Complex Analysis and Differential Geometry

Question No : 11 of 33 **Marks: 1 (Budgeted Time Min**

The definite integral of a complex valued function $w(t) = u(t) + i\,v(t)$ over an interval $a \le t \le b$ is _____ , where u, v are real valued functi of a real variable t.

Answer (Please select your correct option)

- $\int_a^b w(t)dt = \int_a^b u(t)\,v(t)dt$
- $\int_a^b w(t)dt = \int_a^b u(t)dt + i\int_a^b v(t)dt$
- $\int_a^b w(t)dt = \int_a^b u(t)dt - i\int_a^b v(t)dt$
- $\int_a^b w(t)dt = \int_a^b u(t)v(t)dt$

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MTH632 Complex Analysis and Differential Geometry

Question No : 17 of 33

By using the Extension of Cauchy Integral formula $\int_c \frac{f(z)}{(z-z_0)^{n+1}} dz =$ _____.

Answer (Please select your correct option)

- $f^{(n)}(z_0)$
- $2\pi i f^{(n)}(z_0)$
- $\frac{f^{(n)}(z_0)}{n!}$
- $\frac{2\pi i}{n!} f^{(n)}(z_0)$

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MTH632 Complex Analysis and Differential Geometry

Question No : 15 of 33

Let $f(z)=z$ be a continuous complex valued function, then $\int_0^{1+i} f(z)dz =$ _____.

Answer (Please select your correct option)

- 1-i
- 0
- 1+i
- 1

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Question Summary : (Attempted Question)

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MTH632 Complex Analysis and Differential Geometry

Question No : 16 of 33

Using the Cauchy–Goursat theorem when contour C is unit circle $\int_C ze^{-z}dz =$ _____ .

Answer (Please select your correct option)

- i
- -1
- 1
- 0

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A triplet $\bar{q} = (q_1, q_2, q_3)$ is called a _____ of Euclidean 3-space E^3.

nswer (Please select your correct option)

- line
- point

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Answer Saved Successfully!

18

Question Summary : (Attempted Question)

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The circle $z = e^{i\theta}$, $0 \le \theta \le 2\pi$ is centered at _____.

Answer (Please select your correct option)

- (1,1)
- (1,0)
- (0,1)
- (0,0)

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A triplet $\vec{q} = (q_1, q_2, q_3)$ is called a _____ of Euclidean 3-space E^3

Answer (Please select your correct option)

- line
- point

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MTH632 Complex Analysis and Differential Geometry

Question No : 11 of 33

The principal value of 1^{2i} is _____.

Answer (Please select your correct option)

- 0
- 1
- i
- 2i

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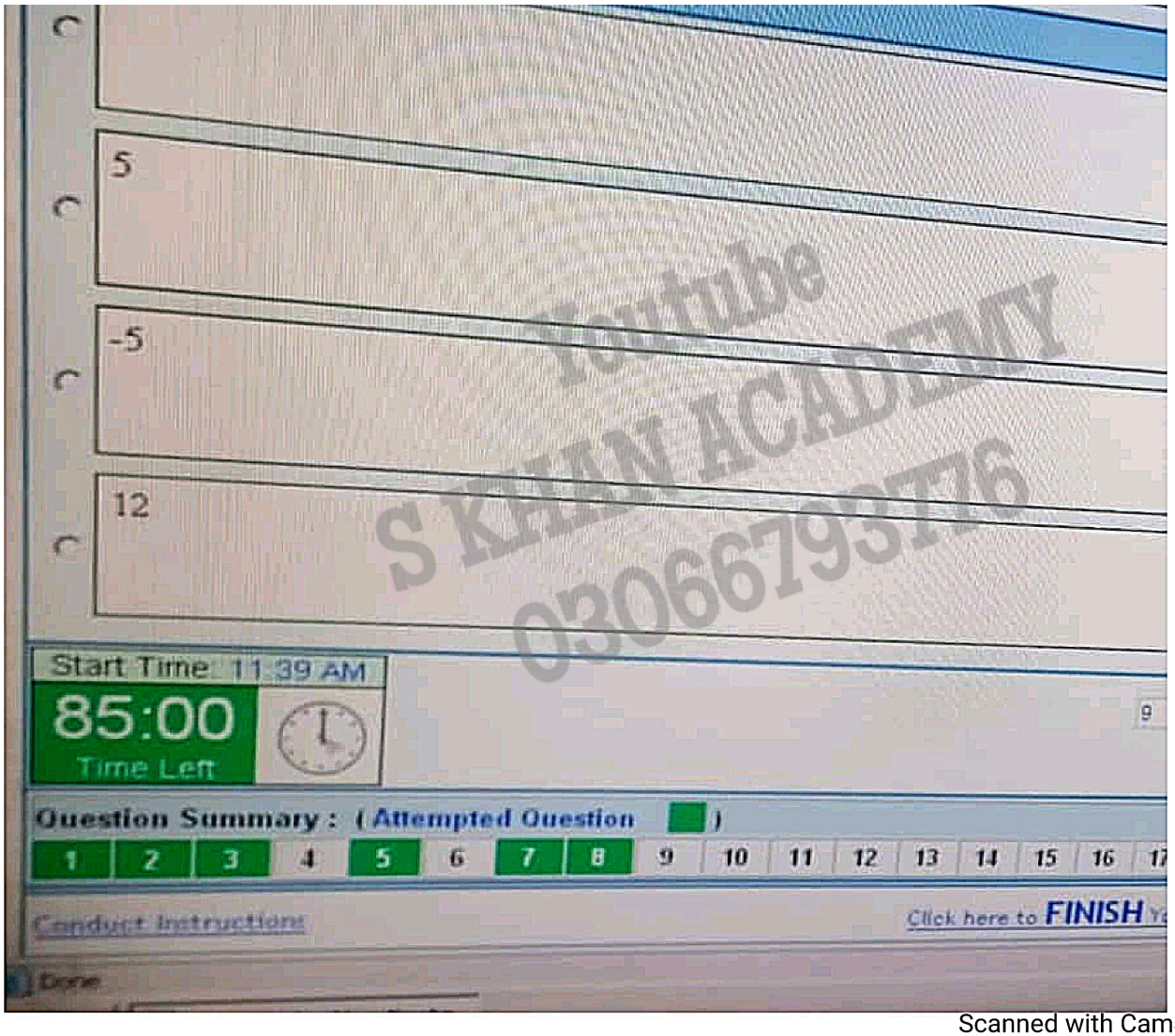
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MTH632 Complex Analysis and Differential Geometry

Question No : 20 of 33 Marks: 1 (B

If the three vectors $\vec{t}$, $\vec{n}$ and $\vec{b}$ are perpendicular to each other and form the Frenet frame, then $\vec{t} \times \vec{n} =$ _____.

Answer (Please select your correct option)

- $-\vec{b}$
- $\vec{b}$
- $\vec{t}$
- $\vec{n}$

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Question Summary (Attempted Question)

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- $\frac{1}{2\pi}\int_C \frac{f(z)}{z-z_0}dz$
- $\int_C \frac{f(z)}{z-z_0}dz$
- $\frac{1}{2\pi}\int_C \frac{f(z)}{z+z_0}dz$
- $\int_C \frac{f(z)}{z+z_0}dz$

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injective

mapping

surjective

bijective

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Question Summary : (Attempted Question)

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Answer (Please select your correct option)

(a) $dx \wedge dx = 0$

(b) $dy \wedge dy = 0$

(c) $dz \wedge dz = 0$

(d) All (a),(b)and(c).

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22

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MTH632 Complex Analysis and Differential Geometry

Question No : 25 of 33

Let $\beta : I \to R^3$ be a unit speed curve. The velocity $T = \beta'$, is said to be the _____ on β.

Answer (Please select your correct option)

- unit
- unit tangent vector field
- vector
- tangent

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Question Summary : (Attempted Question)

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MTH632 Complex Analysis and Differential Geometry

Question No : 7 of 33

A function U is said to be harmonic if and only if _____.

Answer (Please select your correct option)

- $U_{xy} + U_{yx} = 0$
- $U_x + U_y = 0$
- $U_x^2 + U_y^2 = 0$
- $U_{xx} + U_{yy} = 0$

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MTH632 Complex Analysis and Differential Geometry

Question No : 24 of 33

The restriction of a curve $\alpha : I \to R^1$ to a close interval $[a, b]$ is known as a _____.

Answer (Please select your correct option)

- curve
- segment
- line
- curve segment

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Question Summary : (Attempted Question)

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stion No : 21 of 33

he three vectors $\vec{t}$, $\vec{n}$ and $\vec{b}$ are perpendicular to each other and form the Frenet frame, then $\vec{b} \times \vec{t} =$ _____.

swer (Please select your correct option)

- $\vec{b}$
- $\vec{t}$
- $\vec{n}$
- $-\vec{n}$

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Question Summary : (Attempted Question)

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Show that $\arcsin(z) = -i\log(iz + (1-z^2)^{1/2})$.

Answer (Please click here to Add Answer)

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MTH632 Complex Analysis and Differential Geometry

Question No : 8 of 33

The absolute convergence of a series of complex numbers implies the convergence of that _____.

Answer (Please select your correct option)

- sequence
- series

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Question Summary : (Attempted Question)

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Question No : 6 of 33

The function $f(z) = |z|$ is non-constant _____.

Answer (Please select your correct option)

- analytic function
- nowhere analytic function
- non-analytic function
- entire function

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Question No : 32 of 33

Show that

(a) $Log(1+i)^2 = 2Log(1+i)$ and (b) $Log(-1+i)^2 \neq 2Log(-1+i)$

Answer (Please click here to Add Answer)

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Question Summary : (Attempted Question)

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MTH632 Complex Analysis and Differential Geometry

Question No : 5 of 33

A point at which a function ceases to be analytic is called a _____.

Answer (Please select your correct option)

- singular point
- non-singular point
- regular point
- non-regular point

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Question Summary : (Attempted Question)

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Question No :

The sequence $z_n = -2 + i\frac{1}{n^3}$ $(n = 1, 2, 3, \ldots)$ for $n \to \infty$ converges to _____.

Answer (Please select your correct option)

i

-1

1

-2

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Question Summary : (Attempted Question)

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MTH632 Complex Analysis and Differential Geometry

Question No : 2 of 33

The logarithmic function of $e^w = z$, where $z = re^{w}$ and $w = u + iv$, then _____.

Answer (Please select your correct option)

- $w = \ln r + i(\theta + 2n\pi)$
- $w = re^{w}$
- $w = r + i(\theta + 2n\pi)$
- $w = -\ln r + i\,\theta$

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ATH632 Complex Analysis and Differential Geometry

Question No : 1 of 33

In the complex valued function $U(x, y)$ is harmonic in some domain , if _____.

Answer (Please select your correct option)

- $u_{xx} - u_{yy} = 0$
- $u_{xx} + u_{yy} = 0$
- $u_{xx} = u_{yy} = 0$
- $u_{xx} = u_{yy}$

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Question No : 10 of 33

For hyperbolic functions of complex numbers $Cosh\ x$ = _____.

Answer (Please select your correct option)

- ($\frac{e^{x} + e^{-x}}{2i}$
- ($\frac{e^{x} - e^{-x}}{2i}$
- ($\frac{e^{-x} - e^{x}}{2}$
- ($\frac{e^{x} + e^{-x}}{2}$

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Question Summary : (Attempted Question)

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wer (Please select your correct option)

$$\int_a^b w(t)dt = \int_a^b u(t)\,v(t)dt$$

$$\int_a^b w(t)dt = \int_a^b u(t)dt + \int_a^b v(t)dt$$

$$\int_a^b w(t)dt = \int_a^b u(t)dt - \int_a^b v(t)dt$$

$$\int_a^b w(t)dt = \int_a^b u(t)\,v(t)dt$$

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MTH632 Complex Analysis and Differential Geometry

Question No : 6 of 33

The function $f(z) = |z|$ is non-constant _____.

Answer (Please select your correct option)

- analytic function
- nowhere analytic function
- non-analytic function
- entire function

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Question Summary: (Attempted Question)

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1632 Complex Analysis and Differential Geometry

Question No : 4 of 33 Marks: 1 (Budgeted Time Min)

infinite sequence $z_1, z_2, \ldots, z_n, \ldots$ of complex numbers has a limit 'z' if, for each positive number ε, there exists a positive integer n_0 such that $|z_n - z| < \varepsilon$ whenever _____

Answer (Please select your correct option)

- $n = n_0$
- $n < n_0$
- $n > n_0$
- $n \neq n_0$

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Question Summary : (Attempted Question)

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MTH632 Complex Analysis and Differential Geometry

Question No : 22 of 33 **Marks: 1 (Bo**

An important consequence of the wedge product is that "repeats are zero" due to "alternation rule", that means _____

Answer (Please select your correct option)

(a) $dx \wedge dx = 0$

(b) $dy \wedge dy = 0$

(c) $dz \wedge dz = 0$

(d) All (a),(b) and (c)

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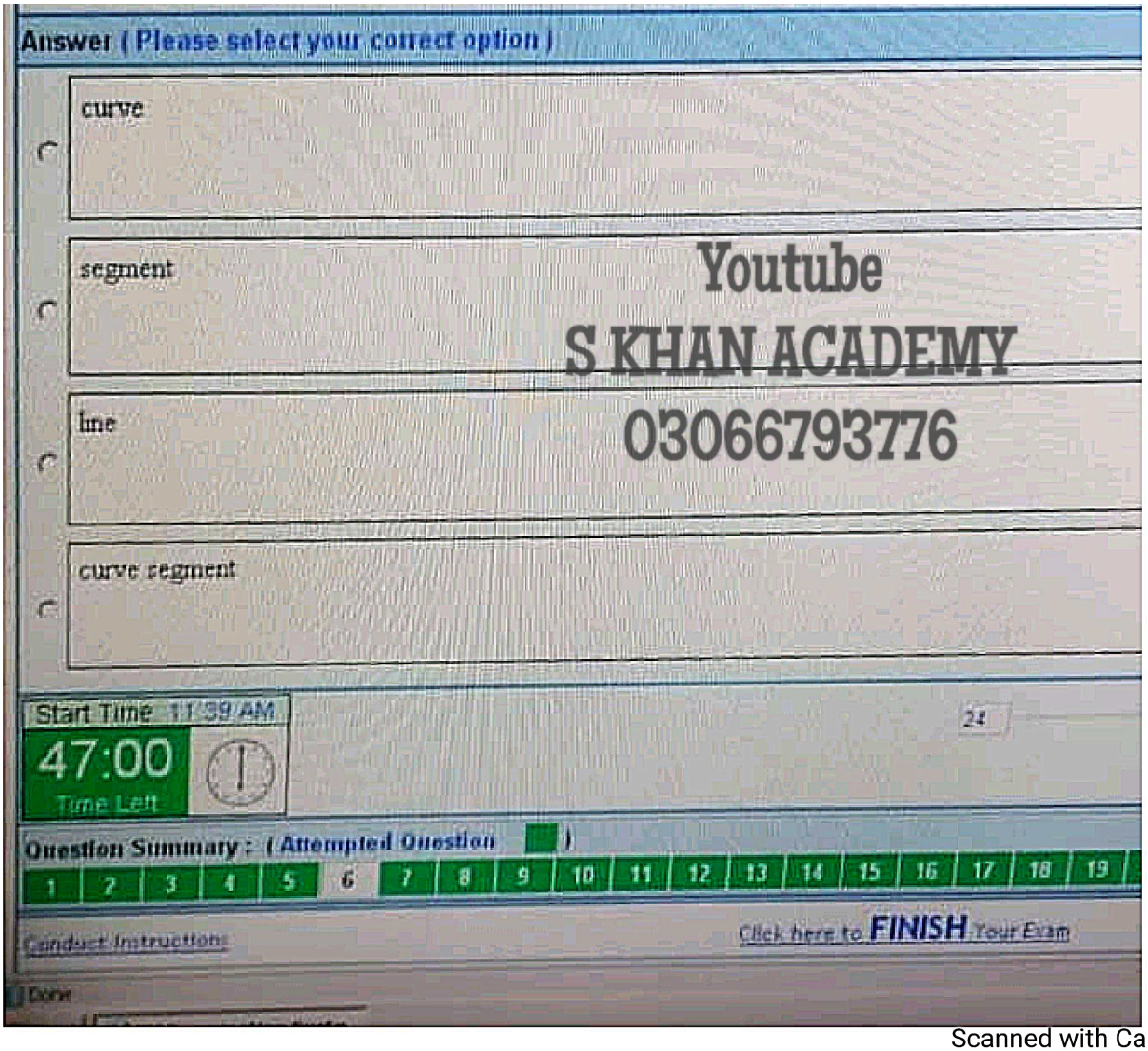
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